Instruction	Name	Format	Opcode	Funct3	Funct7	Description	rs1	rs2	imm	Source Set	ALU Operation	Memory Operation	Result Route	Branch Operation
add	Add	R	0110011	0x0	0x00	rd = rs1 + rs2	rs1	rs2	zero	RS1, RS2	ALU_ADD	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
sub	Subtract	R	0110011	0x0	0x20	rd = rs1 - rs2	rs1	rs2	zero	RS1, RS2	ALU_SUB	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
xor	Exclusive Or	R	0110011	0x4	0x00	rd = rs1 ^ rs2	rs1	rs2	zero	RS1, RS2	ALU_XOR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
or	OR	R	0110011	0x6	0x00	rd = rs1 | rs2	rs1	rs2	zero	RS1, RS2	ALU_OR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
and	And	R	0110011	0x7	0x00	rd = rs1 & rs2	rs1	rs2	zero	RS1, RS2	ALU_AND	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
sll	Shift Left (Logical)	R	0110011	0x1	0x00	rd = rs1 << rs2	rs1	rs2	zero	RS1, RS2	ALU_LSHL	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
srl	Shift Right (Logical)	R	0110011	0x5	0x00	rd = rs1 >> rs2	rs1	rs2	zero	RS1, RS2	ALU_LSHR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
sra	Shift Right (Arithmetic)	R	0110011	0x5	0x20	rd = rs1 >>> rs2	rs1	rs2	zero	RS1, RS2	ALU_ASHR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
slt	Set Less Than	R	0110011	0x2	0x00	rd = (rs1 < rs2) ? 1 : 0	rs1	rs2	zero	RS1, RS2	ALU_SLT	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
sltu	Set Less Than (Unsigned)	R	0110011	0x3	0x00	rd = (rs1 < rs2) ? 1 : 0	rs1	rs2	zero	RS1, RS2	ALU_SLTU	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
addi	Add Immediate	I	0010011	0x0	0x00	rd = rs1 + imm	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
xori	Exclusive Or Immediate	I	0010011	0x4	0x00	rd = rs1 ^ imm	rs1	zero	imm	RS1, IMM	ALU_XOR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
ori	Or Immediate	I	0010011	0x6	0x00	rd = rs1 | imm	rs1	zero	imm	RS1, IMM	ALU_OR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
andi	And Immediate	I	0010011	0x7	0x00	rd = rs1 & imm	rs1	zero	imm	RS1, IMM	ALU_AND	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
slli	Shift Left (Logical) Immediate	I	0010011	0x1	0x00	rd = rs1 << imm[4:0]	rs1	zero	imm	RS1, IMM	ALU_SHL	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
srli	Shift Right (Logical) Immediate	I	0010011	0x5	0x00	rd = rs1 >> imm[4:0]	rs1	zero	imm	RS1, IMM	ALU_SHR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
srai	Shift Right (Arithmetic) Immediate	I	0010011	0x5	0x20	rd = rs1 >>> imm[4:0]	rs1	zero	imm	RS1, IMM	ALU_SHR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
slti	Set Less Than Immediate	I	0010011	0x2	0x00	rd = (rs1 < imm) ? 1 : 0	rs1	zero	imm	RS1, IMM	ALU_SLT	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
sltiu	Set Less Than (Unsigned) Immediate	I	0010011	0x3	0x00	rd = (rs1 < imm) ? 1 : 0	rs1	zero	imm	RS1, IMM	ALU_SLTU	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
lb	Load Byte	I	0000011	0x0	-	rd = {24'M[rs1 + imm][7], M[rs1 + imm][7:0]}	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_READ_ALU	ALU_MEMREAD_RD	BRANCH_NOBRANCH
lh	Load Half	I	0000011	0x1	-	rd = {16'M[rs1 + imm][15], M[rs1 + imm][15:0]}	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_READ_ALU	ALU_MEMREAD_RD	BRANCH_NOBRANCH
lw	Load Word	I	0000011	0x2	-	rd = M[rs1 + imm][31:0]	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_READ_ALU	ALU_MEMREAD_RD	BRANCH_NOBRANCH
lbu	Load Byte (Unsigned)	I	0000011	0x4	-	rd = {24'0, M[rs1 + imm][7:0]}	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_READ_ALU	ALU_MEMREAD_RD	BRANCH_NOBRANCH
lhu	Load Half (Unsigned)	I	0000011	0x5	-	rd = {16'0, M[rs1 + imm][15:0]}	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_READ_ALU	ALU_MEMREAD_RD	BRANCH_NOBRANCH
sb	Store Byte	S	0100011	0x0	-	M[rs1 + imm][7:0] = rs2[7:0]	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_WRITE_ALU	ALU_MEMWRITE	BRANCH_NOBRANCH
sh	Store Half	S	0100011	0x1	-	M[rs1 + imm][15:0] = rs2[15:0]	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_WRITE_ALU	ALU_MEMWRITE	BRANCH_NOBRANCH
sw	Store Word	S	0100011	0x2	-	M[rs1 + imm][31:0] = rs2[31:0]	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_WRITE_ALU	ALU_MEMWRITE	BRANCH_NOBRANCH
beq	Branch If Equal	B	1100011	0x0	-	if (rs1 == rs2) PC += imm	rs1	rs2	imm	RS1, RS2	ALU_XOR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH__OFF_ZERO
bne	Branch If Not Equal	B	1100011	0x1	-	if (rs1 != rs2) PC += imm	rs1	rs2	imm	RS1, RS2	ALU_XOR	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_OFF_NZERO
blt	Branch If Less Than	B	1100011	0x4	-	if (rs1 < rs2) PC += imm	rs1	rs2	imm	RS1, RS2	ALU_SLT	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_OFF_NZERO
bge	Branch If Greater Than or Equal To	B	1100011	0x5	-	if (rs1 >= rs2) PC += imm	rs1	rs2	imm	RS1, RS2	ALU_SLT	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_OFF_ZERO
bltu	Branch If Less Than (Unsigned)	B	1100011	0x6	-	if (rs1 < rs2) PC += imm	rs1	rs2	imm	RS1, RS2	ALU_SLTU	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_OFF_NZERO
bgeu	Branch If Greater Than Or Equal To (Unsigned)	B	1100011	0x7	-	if (rs1 >= rs2) PC += imm	rs1	rs2	imm	RS1, RS2	ALU_SLTU	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_OFF_ZERO
jal	Jump And Link	J	1101111	-	-	rd = PC + 4; PC += imm	zero	zero	imm	RS1, RS2	ALU_ADD	MEM_PASSTHROUGH_ALU	PC_RD_ALU_PC	BRANCH_OFF
jalr	Jump And Link Register	I	1100111	0x0	-	rd = PC + 4; PC = rs1 + imm	rs1	zero	imm	RS1, IMM	ALU_ADD	MEM_PASSTHROUGH_ALU	PC_RD_ALU_PC	BRANCH_ALU
lui	Load Upper Immediate	U	0110111	-	-	rd = imm << 12	zero	zero	imm << 12	RS1, IMM	ALU_ADD	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH
auipc	Add Upper Immediate To Program Counter	U	0010111	-	-	rd = PC + (imm << 12)	zero	zero	imm << 12	PC, IMM	ALU_ADD	MEM_PASSTHROUGH_ALU	ALU_RD	BRANCH_NOBRANCH

Type	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	funct7							rs2					rs1					funct3			rd					opcode						
I	imm[11:0]												rs1					funct3			rd					opcode						
S	imm[11:5]							rs2					rs1					funct3			imm[4:0]					opcode						
B	imm[12]	imm[10:5]						rs2					rs1					funct3			imm[4:1]				imm[11]	opcode						
J	imm[20]	imm[10:1]										imm[11]	imm[19:12]								rd					opcode						
U	imm[31:12]																				rd					opcode						